

2. C.L.Liu and D. P. Mohapatra, "Elements of Discrete Mathematics: A Computer Oriented Approach," Tata McGraw-Hill Edition, 2017.
3. D.S. Malik and M.K. Sen, "Discrete Mathematical Structures: Theory and Applications," Thomson, 2004.
4. Joseph Gallian, "Contemporary Abstract Algebra", 9th Edition, CENGAGE publications, 2019.
5. Gross and Yellen, "Graph Theory and its Applications", 2nd Edition, Chapman and Hall, 2005.
6. Kenneth H. Rosen, "Discrete Mathematics and Its Applications", Sixth Edition, 2006

5. Teaching-Learning Strategies in brief (4 to 5 sentences):

This course supports the expected characteristics, capabilities and skills for computer science graduates in the following ways:

- * Mastery of Computer Science technical foundations
- * Recognition of common Computer Science themes and principles
- * Recognition of interplay between theory and practice
- * Effective problem solving and critical thinking skills

6. Assessment methods and weightages in brief (4 to 5 sentences):

- Assignments: 10%
 - In-Class Tests: 20%
 - Mid Semester Examination: 30%
 - End Semester Examination: 40%
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Title of the Course : Distributed Systems

Faculty Name : Kishore Kothapalli

Course Code : CS3.401

L-T-P : 3-1-0

Credits : 4

(L= Lecture hours, T=Tutorial hours, P=Practical hours)

Name of the Academic Program B. Tech. in Computer Science and Engineering

1. Prerequisite Course / Knowledge:

An understanding of operating systems, networks, and algorithms

2. Course Outcomes (COs):

After completion of this course successfully, the students will be able to..

CO-1: Explain the challenges faced by distributed systems in terms of lack of global time, synchrony, faults, programming support, etc.

CO-2: Employ standard distributed programming frameworks to write distributed programs for problem solving

CO-3: Explain the properties and design principles of various real-world and practical distributed systems

CO-4: Interpret the impact of faults in distributed systems in the context of important problems such as distributed agreement, distributed consensus, and distributed transaction processing

CO-5: Analyze distributed algorithms for graphs with respect to correctness, round complexity, and message complexity.

CO-6: Analyze the limitations of distributed systems and assess the operational scope of large-scale distributed systems

3. Mapping of Course Outcomes (COs) with Program Outcomes (POs) and Program Specific Outcomes (PSOs) – Course Articulation Matrix

	PO 1		PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PS O4
CO1	1		1	1	1	1	1	1	1	1	2	1	2	3	3	3	3
CO 2	1		2	2	2	3	2	1	1	2	1	2	2	3	3	3	3
CO 3	1		1	2	2	1	2	1	1	1	2	1	2	3	3	3	3
CO 4	1		2	2	2	1	2	1	1	2	2	1	2	3	3	3	3
CO 5	2		2	2	2	1	1	1	1	2	1	2	2	3	3	3	3
CO 6	2	2	2	2	2	1	2	1	1	2	2	2	2	3	3	3	3

Note: '3' in the box for 'High-level' mapping, 2 for 'Medium-level' mapping, 1 for 'Low'-level' mapping

4.Detailed Syllabus:

- Unit 1
 - Introduction
 - Communication models
 - Time and Synchronization
 - Practice: MPI/Map-Reduce
- Unit 2
 - Distributed file systems
 - Consensus, Agreement, Locking
 - Practice: GFS, Chubby
- Unit 3
 - Distributed Database systems
 - Practice: NoSQL, MongoDB
- Unit 4
 - Limitations of distributed computing
 - Self-Stabilization
 - CAP Theorem
- Unit 5
 - Distributed algorithms for graphs
 - Advanced Topics such as Blockchain, Distributed Storage, and Distributed Program Verification

Reference Books:

1. A.D. Kshemkalyani, M. Singhal, (2011) Distributed Computing: Principles, Algorithms, and Systems, ISBN: 9780521189842, paperback edition, Cambridge University Press, USA.
2. N. Lynch, 1996. Distributed Algorithms, Morgan Kauffman, USA, Chapter 5.
3. Other significant papers from conferences such as OSDI, USENIX, NSDI, for material that is not part of textbooks

5.Teaching-Learning Strategies in brief:

Lectures of the class use the active learning methodology and allow students to learn concepts thoroughly in class along with practicing small examples. HomeWorks assigned as part of the course are useful to impart knowledge of using practical distributed programming tools and

libraries. To promote teamwork, some of the homework's are done in a team of two students. The overall learning from the course is enhanced by doing a substantial practice-based project – usually in a team of two students. The course will also have a summative assessment in the form of a final/end-semester exam.

6.Assessment methods and weightages in brief:

- In-class Quiz Exams (Cumulative over several): 15%
- HomeWorks: 20%
- Project: 25%
- End Semester Examination: 40%

Title of the course : **Dynamical processes in complex networks**
Faculty Name : Chittaranjan hens, CCNSB
Course Code : **SC1.440**
Credits : **4**
L - T - P : **L=2, T+P=1**
(L - Lecture hours, T-Tutorial hours, P - Practical hours)
Semester, Year : Monsoon 2026
Pre-Requisites: Linear Algebra, Differential Equations, asic of Statistical Physics

Course Outcomes

After completion of the course, the students will be able to..

1. *Interpret and quantify* the connectivity patterns of real-world systems.
2. *Model* how the infection spreads in a well-mixed population setup and how chaos develops in the double pendulum.
3. *Identify* how structure influences synchronization and the spread of disease in metapopulation networks.
4. *Implement* the use of various appropriate pharmaceutical or non-pharmaceutical intervention strategies for reducing the severity of infectious diseases in actual circumstances.
5. *Design* cutting-edge problems regarding the diverse dynamical process in complex networks.

Course Topics :

- A. Networks:** A general introduction. Dynamical processes in networks (1 Lecture)
- B. Fundamentals of Network theory:** Mathematics of networks, Measures and metrics(3 Lectures)
- C. Network Models:** Erdos-Renyi, Scale-free networks, and small-world model. Statistical properties and degree distribution. Random graphs with general degree distributions. Percolation Process in complex networks. (4 Lectures)
- D. Nonlinear Dynamics:** From fixed points to chaos theory. Application: Epidemics (SIR/SIS dynamics) and double pendulum (4 Lectures).
- E. Dynamical Processes in Networks:** Epidemics in complex networks, Role heterogeneity in disease spreading. Pharmaceutical and nonpharmaceutical intervention strategy. Epidemics in metapopulation networks, epidemics in time- varying networks, and epidemics in multiplex networks, Activity-driven networks (4 Lectures). Synchronization in coupled nonlinear oscillators (2 lectures).
- F. Recent literature survey, and problem design** (4 lectures).

Preferred Text Books :